- Midterm 2 is 3 weeks from today

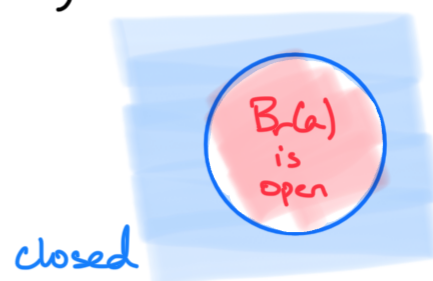
<u>Def</u> Let (X, d) be a metric space. A set $A \subseteq X$ is <u>closed</u> if A^c is open.

<u>Ex</u> Let (X, d) be a metric space.

① $\emptyset$ and X are closed: since $\emptyset^c = X$ and $X^c = \emptyset$ are open.

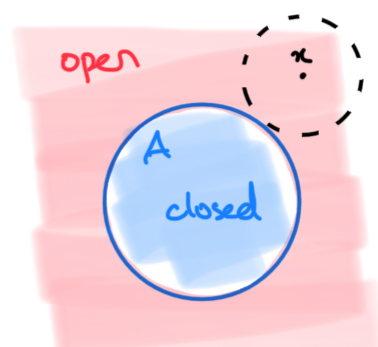
② $(B_r(a))^c = \{x \in X : d(x, a) \geq r\}$ is closed $\forall r > 0$ and $\forall a \in X$: since

$[(B_r(a))^c]^c = B_r(a)$ is open.



③ $A = \{x \in X : d(x, a) \leq r\}$ is closed $\forall r > 0$ and $\forall a \in X$:

Want $A^c = \{x \in X : d(x, a) > r\}$ is open. Fix $x \in A^c$. Then $d(x, a) > r$. Set $s = d(x, a) - r > 0$.



Then $B_s(x) \subseteq A^c$, since

$$
\begin{aligned}
y \in B_s(x) &\Rightarrow d(y, x) < s \\
&\Rightarrow d(a, x) \leq d(a, y) + d(y, x) \\
&\qquad\qquad < d(a, y) + s \\
&\Rightarrow d(a, y) > d(a, x) - s = r
\end{aligned}
$$

$\Rightarrow \quad y \in A^c.$

So A^c is open.

<u>Rmk</u> "closed" $\neq$ "not open". E.g., in $(\mathbb{R}, |\cdot|)$,

- $(0,1)$ is open
- $[0,1]$ is closed
- $\emptyset$ and $\mathbb{R}$ are open and closed
- $(0,1]$ is not open and not closed

<u>Def</u> Let (X,d) be a metric space and $A \subseteq X$. We say:

① $x \in X$ is an <u>adherent point</u> of A if: $\forall r > 0, \quad B_r(x) \cap A \neq \emptyset$.

② $\bar{A} = \{x \in X : x \text{ is an adherent point of } A\}$ is the <u>closure of A</u>.

③ An adherent point x of A is called:

- <u>isolated</u> if $\exists r > 0$ s.t. $B_r(x) \cap A = \{x\}$
- a <u>limit point</u> of A otherwise

④ $A' = \{x \in X : x \text{ is a limit point of } A\}$.

<u>Rmk</u>

- $A \subseteq \bar{A}$ for any set A:

$$x \in A \quad \Rightarrow \quad x \in B_r(x) \cap A \quad \forall r > 0$$

- $x \in A' \iff$ the intersection of $B_r(x)$ and $A \setminus \{x\}$ is nonempty $\forall r > 0$

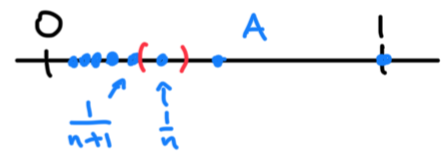
- $\bar{A} = A \cup A'$

<u>Ex</u> $(\mathbb{R}, |\cdot|)$, $A = \{\frac{1}{n} : n \in \mathbb{N}\}$.

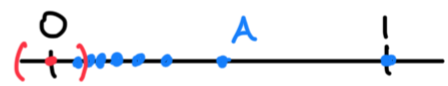
① $\frac{1}{n}$ is an isolated point of A $\forall n \in \mathbb{N}$:

Fix n. Set $r = \frac{1}{n} - \frac{1}{n+1} > 0$.

Then $B_r(\frac{1}{n}) \cap A = \{\frac{1}{n}\}$.



② 0 is a limit point for A:



Fix $r > 0$. Pick $n \in \mathbb{N}$ s.t. $\frac{1}{n} < r$.

Then $\frac{1}{n} \in (-r, r) \cap A$, so $(-r, r) \cap A \neq \emptyset$.

③ $\bar{A} = A \cup \{0\}$

<u>Lem</u> Let (X, d) be a metric space and $A \subseteq X$.

① $(\bar{A})^c = (A^c)^\circ$

② $(A^\circ)^c = \overline{A^c}$

③ $A = \bar{A} \iff A$ is a closed set.

<u>Pf</u>: ① $x \in (\bar{A})^c \iff x \notin \bar{A}$

$\iff \exists r > 0$ s.t. $B_r(x) \cap A = \emptyset$

$\iff \exists r > 0$ s.t. $B_r(x) \subseteq A^c$

$\iff x \in (A^c)^\circ$

② Apply ① to A^c:

$$(\overline{A^c})^c = A^\circ \implies \overline{A^c} = (A^\circ)^c$$

③ A closed $\iff A^c$ open $\iff \underbrace{(A^c)^\circ}_{= (\bar{A})^c \text{ by ①}} = A^c$

$\iff \quad \bar{A} = A. \qquad \square$

<u>Exc</u> Let (X, d) be a metric space and $A, B \subseteq X$.

① If $A \subseteq B$ then $\bar{A} \subseteq \bar{B}$.

② $\overline{A \cap B} \subseteq \bar{A} \cap \bar{B}$

③ $\bar{A} \cup \bar{B} = \overline{A \cup B}$

④ $\bar{\bar{A}} = \bar{A}$. In particular, $\bar{A}$ is a closed set.

⑤ $\bar{A}$ is the smallest closed set containing A.

⑥ A finite union of closed sets is a closed set.

⑦ An arbitrary intersection of closed sets is a closed set.

- Use the proposition from last lecture and take the complement of each statement.
- E.g., for ①,

$$A \subseteq B \implies A^c \supseteq B^c \implies (A^c)^\circ \supseteq (B^c)^\circ$$
$$\implies \bar{A}^c \supseteq \bar{B}^c \implies \bar{A} \subseteq \bar{B}$$

SUBSPACES OF A METRIC SPACE

<u>Def</u> Let (X, d_X) be a metric space and let $\emptyset \neq Y \subseteq X$. Then $d_Y : Y \times Y \to \mathbb{R}$, $d_Y(x,y) = d_X(x,y)$ is a metric on Y, called the <u>induced metric</u> on Y. The metric space (Y, d_Y) is called

a <u>subspace</u> of (X, d_X).

<u>Ex</u> $\mathbb{R}$ with $d_{\mathbb{R}}(x,y) = |x-y|$ is a metric space.

$[0,1]$ with $d_{[0,1]}(x,y) = |x-y|$ is a subspace.